

Keynote Speaker

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The Hitchhiker's Guide to the Metaverse

ABSTRACT

We envision in the future the virtual world will mix and co-exist with the physical world in such an immersive way that we cannot tell what is real and what is virtual. We will live and interact with the virtual objects that are blended into our environments with advanced holographic technology or with high-quality head mounted displays and lose the virtuality boundary. We call such a new reality Surreality. Our vision of “metaverse” is a multi-world. There are multiple virtual worlds developed by different technology companies and there is also the Surreality where real and virtual merged. While the metaverse may seem futuristic, catalysed by emerging technologies such as Extended Reality, 5G, and Artificial Intelligence, the digital “big bang” of our cyberspace is not far away. This talk aims to offer a comprehensive framework that examines the latest metaverse development under the dimensions of state-of-the-art technologies and metaverse ecosystems, illustrates the possibility of the digital “big bang”, and propose a concrete research agenda for the development of the metaverse. Reality will die; long live Surreality.

REFERENCES

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BIOGRAPHY

Pan Hui is a Professor of Computational Media and Arts and Director of the Center for Metaverse and Computational Creativity at the Hong Kong University of Science and Technology. He is also the Nokia Chair in Data Science at the University of Helsinki. He received his PhD from the Computer Laboratory at University of Cambridge, and both his Bachelor and MPhil degrees from the University of Hong Kong. He was an adjunct Professor of social computing and networking at Aalto University, Finland, and a Distinguished Scientist at the Deutsche Telekom Laboratories (T-labs), Germany. His industrial profile also includes his research at Intel Research Cambridge, UK and Thomson Research Paris, France.

Pan Hui is a world-leading expert in Augmented Reality and Mobile Computing, with more than 400 research papers, 30 patents, and over 22,000 citations. He is an International Fellow of the Royal Academy of Engineering, a member of Academia Europaea, an IEEE Fellow, and an ACM Distinguished Scientist. He has founded and chaired many IEEE/ACM conferences/workshops, and has served as track chair, senior program committee member, organising committee member, and program committee member of numerous top conferences including ACM WWW, ACM SIGCOMM, ACM Mobisys, ACM MobiCom, ACM CoNext, IEEE Infocom, IEEE PerCom, IEEE ICNP, IEEE ICDCS, IJCAI, AAAI, UAI, and ICWSM. He served as an Associate Editor for IEEE Transactions on Mobile Computing and IEEE Transactions on Cloud Computing, and as guest editor for various journals including IEEE Journal on Selected Areas in Communications (JSAC) and IEEE Transactions on Secure and Dependable Computing. He also served on the IEEE Computer Society Fellow Evaluation Committee.